

# Experiment: Prolog Program to Find the Sum of Integers from 1 to N

## AIM

To write and execute a **Prolog program to find the sum of integers from 1 to N** using recursion.

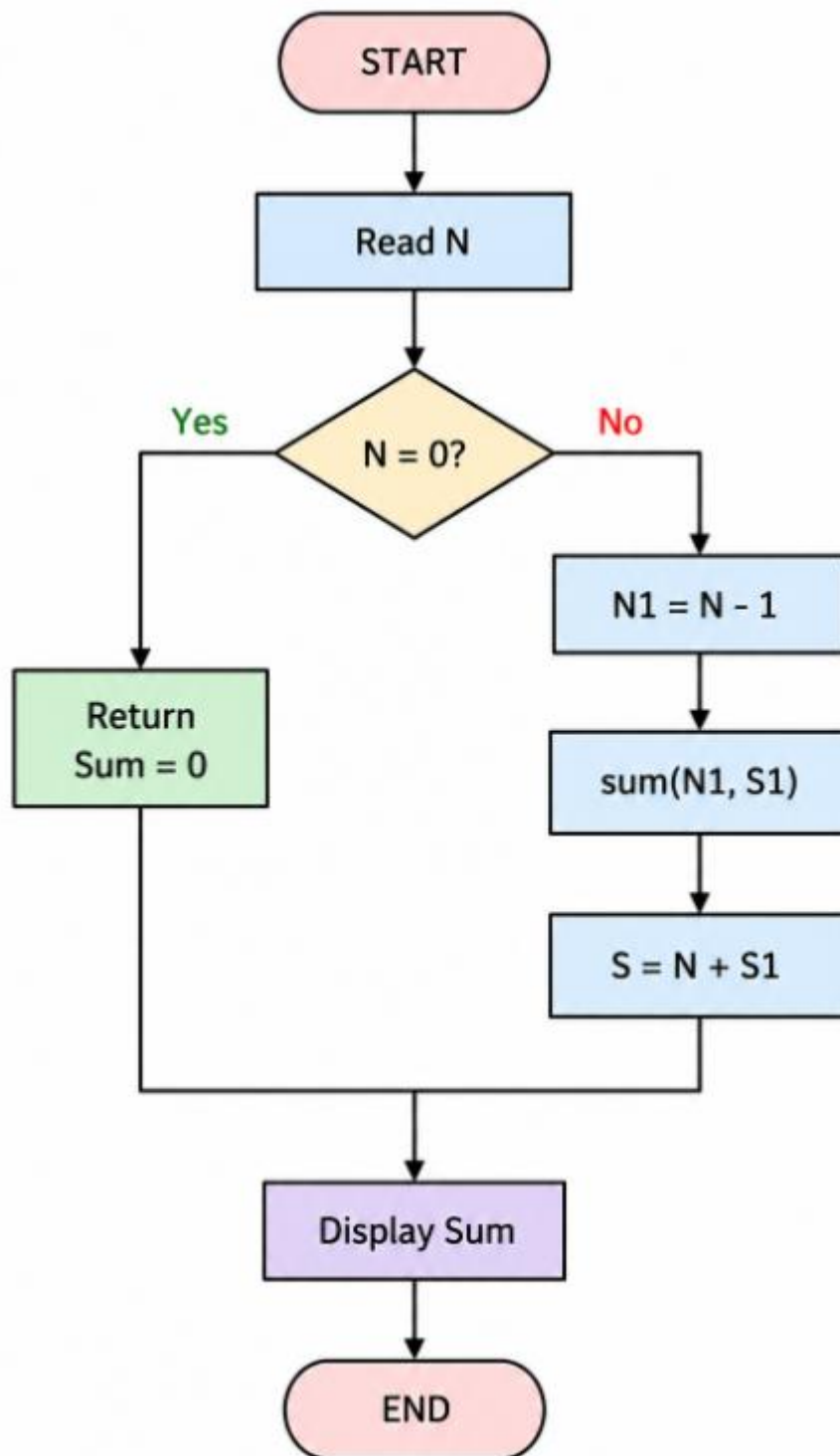
## OBJECTIVE

- To understand recursion in Prolog.
- To implement arithmetic operations in Prolog.
- To calculate the sum of integers from 1 to N.

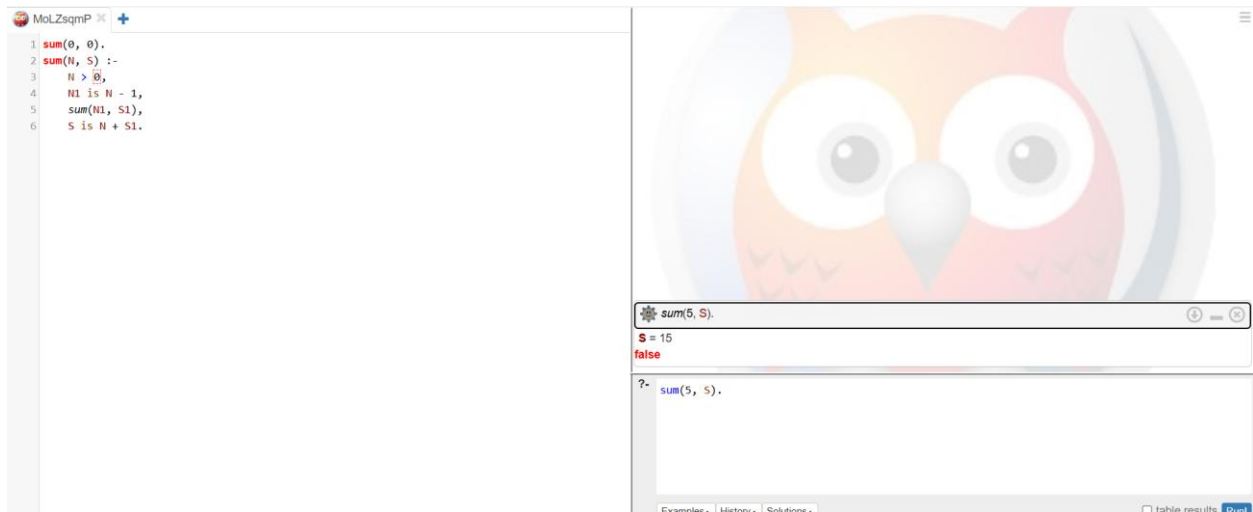
## ALGORITHM

1. Start.
2. Read the value of N.
3. If  $N = 0$ , return sum as 0.
4. Otherwise, calculate  $N - 1$ .
5. Recursively find the sum from 1 to  $N - 1$ .
6. Add N to the recursive result.
7. Display the final sum.
8. Stop.

## FLOWCHART



Code and output:



## RESULT

The Prolog program was successfully executed to calculate the sum of integers from **1 to N**. For  $N = 5$ , the obtained sum is **15**.

## CONCLUSION

Thus, the Prolog program successfully demonstrates the use of **recursion and arithmetic operations** to find the sum of integers from 1 to N.